Patent Intelligence Platform | Relational model

Sheet 1 of 2 - Patent catalog and knowledge relationships

Mapping from Chen sheet 1 | Full table attributes, primary/foreign keys, and junction tables

domains

KEY	ATTRIBUTE	SQL TYPE
PK	**domain_id**	INT
UQ	name	VARCHAR(100)

patent_domains

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**domain_id**	INT

All domain memberships (M:N).

patent_cpc_codes

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**cpc_code**	VARCHAR(32)

CLASSIFIED AS (M:N).

cpc_codes

KEY	ATTRIBUTE	SQL TYPE
PK	**cpc_code**	VARCHAR(32)
	section	CHAR(1)
	description ?	TEXT

patents

KEY	ATTRIBUTE	SQL TYPE
PK	**patent_id**	VARCHAR(80)
	title	TEXT
	abstract	TEXT
	publication_year ?	SMALLINT
	legal_status ?	VARCHAR(50)
	jurisdiction ?	VARCHAR(10)
	cites_patent_count ?	INT
	family_size ?	INT
	cited_by_patent_count	INT
	url ?	VARCHAR(2048)
FK	domain_id ?	INT
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

Year: 1800-2200 when known.
Current incoming citation count is nonnegative.

assignees

KEY	ATTRIBUTE	SQL TYPE
PK	**assignee_id**	INT
UQ	name	VARCHAR(500)

patent_assignees

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**assignee_id**	INT

ASSIGNED TO (M:N).

patent_npl_references

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**npl_id**	BIGINT

CITES (M:N).

npl_references

KEY	ATTRIBUTE	SQL TYPE
PK	**npl_id**	BIGINT
	citation	TEXT
UQ	normalized_key	CHAR(64)

inventors

KEY	ATTRIBUTE	SQL TYPE
PK	**inventor_id**	INT
UQ	name	VARCHAR(500)

patent_inventors

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**inventor_id**	INT

INVENTS (M:N).

Optional preferred display domain (separate from all-domain memberships)

citation_snapshots

KEY	ATTRIBUTE	SQL TYPE
PK	**snapshot_id**	BIGINT
FK	patent_id	VARCHAR(80)
	cited_by_count	INT
	observed_on	DATETIME
	created_at	TIMESTAMP

UQ (patent_id, observed_on)
HAS: one patent, zero or more snapshots.

patent_families

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**patent_id**	VARCHAR(80)
PK/FK	**related_patent_id**	VARCHAR(80)
PK	**relation_type**	VARCHAR(32)

RELATED TO: both FK fields reference patents.
A publication cannot be related to itself.

Chen relationships become relational keys

ASSIGNED TO, INVENTS, CLASSIFIED AS and CITES become separate junction tables.
patent_domains retains every domain membership; patents.domain_id selects a preferred domain.
The family relation has a three-part PK: patent_id, related_patent_id and relation_type.
Citation snapshots keep observation history; patent publication IDs remain separate records.

PK = primary key (underlined) FK = foreign key UQ = unique constraint ? = NULL permitted

All PK-marked fields in a table form ONE key. Arrows point from FK fields to referenced PKs; joined branches share that PK target.

Source: 01_schema.sql + 05_auth.sql

Defaults and secondary indexes omitted. [ref] repeats an existing table.

Patent Intelligence Platform | Relational model

Sheet 2 of 2 - Accounts, saved analysis and authentication

Mapping from Chen sheet 2 | Authentication support tables added from the current implementation

app_users

KEY	ATTRIBUTE	SQL TYPE
PK	**user_id**	CHAR(36)
UQ	email	VARCHAR(320)
	display_name	VARCHAR(120)
	password_hash ?	VARCHAR(255)
	is_active	BOOLEAN
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

user_roles

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**user_id**	CHAR(36)
PK/FK	**role_id**	INT

HAS ROLE (M:N).

roles

KEY	ATTRIBUTE	SQL TYPE
PK	**role_id**	INT
UQ	name	VARCHAR(50)

patents [ref]

KEY	ATTRIBUTE	SQL TYPE
PK	**patent_id**	VARCHAR(80)

Same publication table as sheet 1.

Cases have zero or more runs. Each saved result references one run and one patent.

An evaluation is optional. Multiple recommendations may belong to one run.

auth_sessions

KEY	ATTRIBUTE	SQL TYPE
PK	**token_hash**	CHAR(64)
FK	user_id	CHAR(36)
	expires_at	DATETIME
	created_at	TIMESTAMP

Stores token hashes, never raw session tokens.

analysis_cases

KEY	ATTRIBUTE	SQL TYPE
PK	**case_id**	CHAR(36)
FK	owner_user_id ?	CHAR(36)
	title	VARCHAR(250)
	idea_text	TEXT
	status	ENUM
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

Owner may be NULL for preserved legacy cases.

analysis_runs

KEY	ATTRIBUTE	SQL TYPE
PK	**run_id**	CHAR(36)
FK	case_id	CHAR(36)
	query_id ?	VARCHAR(64)
	idea_text ?	TEXT
	gnn_mode	ENUM
	top_k	TINYINT unsigned
	run_status	ENUM
	started_at	DATETIME
	completed_at ?	DATETIME
	pipeline_payload ?	JSON
	improvement_payload ?	JSON

query_id is NOT unique; run_id identifies a run.
top_k: 1-100. Each run belongs to one case.

run_patent_results

KEY	ATTRIBUTE	SQL TYPE
PK/FK	**run_id**	CHAR(36)
PK/FK	**patent_id**	VARCHAR(80)
	rank_position	SMALLINT unsigned
	source	ENUM
	expansion_type ?	VARCHAR(32)
	semantic_score ?	DOUBLE
	graph_score ?	DOUBLE
	combined_score ?	DOUBLE
	novelty_score ?	DOUBLE

UQ (run_id, rank_position)

auth_rate_limits

KEY	ATTRIBUTE	SQL TYPE
PK	**bucket_key**	CHAR(64)
	attempts	INT
	window_started	DATETIME

Independent hashed throttle buckets.
No foreign key to users or sessions.

Chen relationship mapping

HAS ROLE -> user_roles (composite PK).
OWNS -> analysis_cases.owner_user_id.
CONTAINS -> analysis_runs.case_id.
RETRIEVES RESULT -> run_patent_results.
EVALUATES -> evaluation_metrics.run_id (UNIQUE).
PRODUCES -> improvement_recommendations.run_id.

evaluation_metrics

KEY	ATTRIBUTE	SQL TYPE
PK	**metric_id**	BIGINT
FK/UQ	run_id	CHAR(36)
	patentability_score	DOUBLE
	risk	VARCHAR(20)
	verdict	VARCHAR(100)
	payload	JSON
	created_at	TIMESTAMP

UNIQUE run_id: zero or one evaluation per run.

improvement_recommendations

KEY	ATTRIBUTE	SQL TYPE
PK	**recommendation_id**	BIGINT
FK	run_id	CHAR(36)
	position	SMALLINT unsigned
	category	VARCHAR(80)
	recommendation	TEXT
	rationale ?	TEXT

UQ (run_id, position)

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All PK-marked fields in a table form ONE key. Arrows point from FK fields to referenced PKs; joined branches share that PK target.

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